

DUO ZHANG

Ph.D. Candidate in Computer Science, Rutgers University

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Research: Motion planning and manipulation, with an emphasis on high-performance geometric and sampling-based algorithms, trajectory optimization, GPU acceleration, and real-robot systems.

Education

Rutgers University

September 2023–Present

Ph.D. Candidate, Computer Science. Advisor: Prof. Jingjin Yu.

New York University

January 2021–December 2022

M.S., Courant Institute of Mathematical Sciences.

Shandong University

September 2016–June 2020

B.E., Computer Science; prior study in Energy and Power Engineering (2016–2018).

Selected honor: National Scholarship, Ministry of Education of China.

Research Experience

Rutgers University, Algorithmic Robotics & Control Lab

September 2023–Present

Graduate Assistant | Advisor: Prof. Jingjin Yu

- **Geometric and online motion planning (RVG / dRVG).** Developed rotation-stacked visibility graphs (RVG) for resolution-complete, asymptotically optimal multi-query planning of polygonal robots. Extended RVG to online planning in unknown static environments using incremental graph merging and quadtree-guided exploration (dRVG); validated online navigation on microMVP. [2], [5]
- **Dual-arm task and motion planning (SDAR).** Integrated dependency-driven task planning with synchronous dual-arm motion planning, exploiting tabletop rearrangement structure. Achieved **100% end-to-end success** on evaluated non-monotone rearrangement tasks in simulation; validated on dual UR5e arms. [1]
- **GPU-parallel space-time planning (ST-pRRTC).** Developed a GPU-parallel space-time RRT-Connect planner with a shared forward tree and adaptive goal-time forests for known moving-obstacle trajectories; demonstrated UR5e motion among moving Crazyflie quadrotors. [6]
- **Free-floating space manipulation (FAVOR).** Developed Bézier joint-velocity optimization with analytical recursive sensitivities for reaching, tracking, and prescribed-time interception under coupled spacecraft–arm dynamics. Achieved **99.8% point-to-point success** at **1.78 s mean computation** in simulation with a seven-DoF arm and five spacecraft models; timing includes failures and timeouts. [7]

Tencent America, Lightspeed & Quantum Studio

June 2022–June 2023

Research & Development Intern, Copernicus Research Group

Bellevue, WA

- **Collision-constrained trajectory optimization.** Developed a semi-infinite programming solver with conservative motion bounds and adaptive subdivision to maintain feasibility; evaluated on robot arms and UAVs. [3]

New York University, Courant Institute

September 2021–December 2022

Research Assistant with Profs. Daniele Panozzo and Lerrel Pinto

New York, NY

- **Deformable-object manipulation.** Integrated PolyFEM simulation with reinforcement-learning environments; built camera-based tracking and validated physical robot experiments, including a silicone-ball slingshot.

Shandong University, Interdisciplinary Research Center

January–December 2020

Research Assistant, Prof. Changhe Tu’s lab

Qingdao, China

- **Power grasp planning.** Worked with Xifeng Gao and Zherong Pan on infinite programming with complementary constraints, kernel-integral relaxation, and fast multipole evaluation for optimization-based grasp planning. Published in *IEEE RA-L*. [4]

University of California, Los Angeles

July–September 2019

Research Assistant, Prof. Sriram Sankararaman’s lab

Los Angeles, CA

- **Cell-type-specific gene expression prediction.** Developed tensor-component-analysis methods for association studies and published an R package through UCLA’s Cross-disciplinary Scholars program.

Publications and Preprints

Conference and journal publications

- [1] **Duo Zhang**, Junshan Huang, and Jingjin Yu. “High-Performance Dual-Arm Task and Motion Planning for Tabletop Rearrangement.” *IEEE ICRA*, 2026. [Paper](#).
- [2] **Duo Zhang**, Zihe Ye, and Jingjin Yu. “Asymptotically-Optimal Multi-Query Path Planning for a Polygonal Robot.” *IEEE ICRA*, 2025. [Paper](#).
- [3] **Duo Zhang**, Chen Liang, Xifeng Gao, Kui Wu, and Zherong Pan. “Provably Feasible Semi-Infinite Program Under Collision Constraints via Subdivision.” *IEEE Transactions on Robotics (T-RO)*, 2024. [Paper](#).
- [4] Zherong Pan, **Duo Zhang**, Changhe Tu, and Xifeng Gao. “Planning of Power Grasps Using Infinite Program Under Complementary Constraints.” *IEEE Robotics and Automation Letters (RA-L)*, 7(1):650–657, 2022. [Paper](#).

Preprints

- [5] **Duo Zhang**, Hechen Zhang, Junshan Huang, and Jingjin Yu. “dRVG: Quadtree-Guided, Resolution-Complete Online Motion Planning for Polygonal Robots in Unknown Environments.” *arXiv preprint*, 2026. [arXiv:2609.31412](#).
- [6] **Duo Zhang**, Jintong Li, Junshan Huang, and Jingjin Yu. “ST-pRRTC: Parallel Space-Time RRT-C with Adaptive Goal-Time Forests.” *arXiv preprint*, 2026. [arXiv:2609.30533](#).
- [7] **Duo Zhang**, Zhizhuo Zhang, Xiaoli Bai, and Jingjin Yu. “Efficient Bézier Velocity Optimization for Free-Floating Space Manipulators.” *arXiv preprint*, 2026. [arXiv:2609.31880](#).

Teaching Experience

Teaching Assistant, Rutgers University

- Software Methodology Fall 2024, Spring 2025
- Data Management for Data Science Spring 2024
- Systems Programming Fall 2023